

CENTRE FOR ADVANCED STUDIES
DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, UTTAR PRADESH,
LUCKNOW



EVALUATION SCHEME AND SYLLABI

For

B.Tech. 2nd Year
Mechanical & Mechatronics Engineering
(Additive Manufacturing)

Effective from the Session: 2026-27

DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY UTTAR PRADESH.
Sec-11, Jankipuram Vistar, Lucknow-226031, Uttar Pradesh, India

DETAILED SYLLABI

B.Tech. Second Year

Effective-from Session 2026-27

B.Tech. (Mechanical & Mechatronics Engineering (Additive Manufacturing))
Second Year, Semester-III

								Evaluation Scheme					
SN	Subject Code	Subject Name	Type	Category	Period			Sessional Component		Sessional (SW)(TS/PS)	End Semester Examination (ESE)	Total	Credit
					L	T	P	CT	TA	CT+TA	TE/PE	SW+ESE	Cr
1.	CAS 302	Engineering Mathematics-III	T	BS	3	0	0	20	10	30	70	100	3
2.	CME 301	Fundamentals of Thermodynamics and Fluid Mechanics	T	ES	3	1	0	20	10	30	70	100	4
3.	CME 302	Materials Science & Metallurgy	T	ES	3	0	0	20	10	30	70	100	3
4.	CME 303	Introduction to Robotics	T	ES	3	0	0	20	10	30	70	100	3
5.	CME 304	Analog and Digital Electronics	T	ES	3	1	0	20	10	30	70	100	4
6.	CME 351	Thermodynamics and Fluid Lab	T	ES	0	0	2	-	50	50	50	100	1
7.	CME 353	Robotics Lab	P	ES	0	0	2	-	50	50	50	100	1
8.	CME 354	Analog and Digital Electronics Lab	P	ES	0	0	2	-	50	50	50	100	1
9.	CVA 301	Constitution of India*	P	VA	2*	0	0	20	10	30*	70*	100*	1
					15+2*	2	6			300+ 30*	500+70*	800+100*	21
10.	Vocational Training / Skill based courses (Mandatory for Exit after II year)				-	-	-	-	-	-	-	-	2

*Compulsory Qualifying Audit Course

Abbreviation Used:

BS: Basic Science Course

ES: Engineering Science Course

VA: Value Added Course

B.Tech. Second Year, Semester-IV: Evaluation Scheme
B.Tech. Mechanical & Mechatronics Engineering (Additive Manufacturing)

								Evaluation Scheme					
SN	Subject Code	Subject Name	Type	Category	Period			Sessional Component		Sessional (SW)(TS/PS)	End Semester Examination (ESE)	Total	Credit
					L	T	P	CT	TA	CT+TA	TE/PE	SW+ESE	Cr
1.	CME 401	Mechanics of Solids	T	ES	3	1	0	20	10	30	70	100	4
2.	CME 402	Manufacturing Science	T	ES	3	0	0	20	10	30	70	100	3
3.	CME 403	Computer Aided Design and Additive Manufacturing	T	ES	3	0	0	20	10	30	70	100	3
4.	CME 404	Introduction to Automation	T	ES	3	0	0	20	10	30	70	100	3
5.	CME 405	Introduction to Artificial Intelligence	T	ES	3	0	0	20	10	30	70	100	3
6.	CME 451	Materials Testing Lab	P	ES	0	0	2	-	50	50	50	100	1
7.	CME 452	Manufacturing and Automation Lab	P	ES	0	0	2	-	50	50	50	100	1
8.	CME 453	Computer Aided Design and Additive Manufacturing Lab	P	ES	0	0	2	-	50	50	50	100	1
9.	CME 456	Mini Project	P	ES	0	0	2	-	50	50	50	100	1
10.	CVA 401	Universal Human Values*	T	VA	2*	0	0	20	10	30*	70*	100*	1
					15+2*	1	08			350+30*	550+70*	900+100*	21
10.	Vocational Training / Skill based courses (Mandatory for Exit after II year)				-	-	-	-	-	-	-	-	2

* Compulsory Qualifying Audit Course

Abbreviation Used:

BS: Basic Science Course

ES: Engineering Science Course

VA: Value Added Course

Summer Internship (4-week)/NPTEL Course (4 week) during summer break after Semester-II and same will be assessed / evaluated in the Semester-III

B.Tech. Mechanical & Mechatronics Engineering (Additive Manufacturing)
Second Year, Semester-III

CAS 302: ENGINEERING MATHEMATICS-III

Course Objectives:

The objective of this course is to familiarize the prospective engineers with techniques in ordinary differential equations, Laplace transform, statistical techniques. It aims to equip the students to deal with advanced level of mathematics and applications essential for their disciplines.

1. The effective mathematical tools for the solutions of differential equations that model physical processes.
2. The basic knowledge of Laplace transforms and its applications in solving differential equations.
3. The tool for convergence of Statistical Techniques for learning advanced Engineering Mathematics.

Syllabus:

Content	Contact Hours
Unit-1: Partial Differential Equations	8
Origin of Partial Differential Equations, Linear and Non-Linear Partial Equations of first order, Lagrange's Equations, Charpit's method, Cauchy's method of Characteristics, Solution of Linear Partial Differential Equation of Higher order with constant coefficients, Equations reducible to linear partial differential equations with constant coefficients.	
Unit-2: Applications of Partial Differential Equations	8
Classification of linear partial differential equation of second order, Method of separation of variables, Solution of wave and heat conduction equation up to two dimension, Laplace equation in two dimensions, Equations of Transmission lines.	
Unit-3: Statistical Techniques I	8
Introduction: Measures of central tendency, Moments, Moment generating function (MGF), Skewness, Kurtosis, Curve Fitting, Method of least squares, Fitting of straight lines, Fitting of second degree parabola, Exponential curves, Correlation and Rank correlation, Regression Analysis: Regression lines of y on x and x on y, regression coefficients, properties of regression coefficients and non-linear regression.	
Unit-4: Statistical Techniques II — Probability and Distribution	8
Introduction, Addition and multiplication law of probability, Conditional probability, Baye's theorem, Random variables (Discrete and Continuous), Probability mass function and Probability density function, Expectation and variance, Discrete and Continuous Probability distributions: Binomial, Poisson and Normal distributions.	
Unit-5: Statistical Techniques III — Sampling, Testing of Hypothesis and Statistical Quality Control	8
Introduction, Sampling Theory (Small and Large), Hypothesis, Null hypothesis, Alternative hypothesis, Testing a Hypothesis, Level of significance, Confidence	

limits.	
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Course Outcomes:

Course Outcome (CO)		Bloom's Level
At the end of this course, the students will be able to:		
CO 1	Solve linear and non-linear first-order PDEs using Lagrange's, Charpit's, and Cauchy's methods, and solve higher-order PDEs with constant coefficients.	K3
CO 2	Apply the method of separation of variables to solve wave, heat, and Laplace equations, and classify second-order PDEs by type.	K3
CO 3	Compute statistical measures (moments, skewness, kurtosis), fit curves using least squares, and determine correlation and regression for engineering data.	K3
CO 4	Apply laws of probability, Bayes' theorem, and standard probability distributions (Binomial, Poisson, Normal) to model and solve engineering uncertainty problems.	K3
CO 5	Apply sampling theory and hypothesis testing (z-test, t-test, chi-square-test) to draw statistical inferences and implement statistical quality control.	K3

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate, K6 – Create

Text Books:

- B. V. Ramana, Higher Engineering Mathematics, Tata McGraw-Hill Publishing Company Ltd., 2008.
- B. S. Grewal, Higher Engineering Mathematics, Khanna Publisher, 2005.
- R. K. Jain & S. R. K. Iyenger, Advance Engineering Mathematics, Narosa Publishing House, 2002.

Reference Books:

- E. Kreyszig, Advance Engineering Mathematics, John Wiley & Sons, 2005.
- Peter V. O'Neil, Advance Engineering Mathematics, Thomson (Cengage) Learning, 2007.
- Maurice D. Weir, Joel Hass, Frank R. Giordano, Thomas, Calculus, Eleventh Edition, Pearson.
- P. G. Hoel, S. C. Port and C. J. Stone, Introduction to Probability Theory, Universal Book Stall, 2003.
- W. Feller, An Introduction to Probability Theory and its Applications, Vol. 1, 3rd Ed., Wiley, 1968.

CME 301: FUNDAMENTALS OF THERMODYNAMICS AND FLUID MECHANICS

Course Objectives:

The objective of this course is to provide students with a solid foundation in thermodynamics and fluid mechanics principles essential for mechanical and mechatronic engineering.

1. Understand thermodynamic systems, laws of thermodynamics and their applications in engineering.
2. Apply concepts of entropy, availability and thermodynamic cycles to engineering problems.
3. Study properties of fluids and analyze fluid statics for engineering applications.
4. Analyze fluid flow equations to solve problems involving pipes and fluid machinery.
5. Apply thermofluidic concepts to mechanical and mechatronic systems.

Syllabus:

Content	Contact Hours
Unit-1: Thermodynamic Fundamentals	8
Thermodynamic systems, Properties of pure substances, State, process, cycle, Zeroth law and temperature measurement, Energy interactions: heat and work in the process, First law of thermodynamics, Internal energy, enthalpy.	
Unit-2: Second Law and Entropy	8
Limitations of first law; Second law of thermodynamics; Heat engines, refrigerators, heat pumps; Carnot cycle; Entropy concept; Entropy change for simple processes; Availability and irreversibility; Basic thermodynamic cycles; Thermodynamic interactions in manufacturing processes.	
Unit-3: Properties of Fluids and Fluid Statics	8
Fluid, Fluid properties; Density, viscosity, surface tension; Compressibility; Significance of Fluid properties in different manufacturing systems and mechanical processes; Pressure measurement; Hydrostatic law; Manometers; Buoyancy and flotation; Stability of floating bodies.	
Unit-4: Fluid Kinematics and Dynamics	8
Types of fluid flow: Streamline, pathline, streakline; Continuity equation; Bernoulli's equation; Euler's equation; Momentum equation; Applications in pipes and ducts; Measurement of flow rate.	
Unit-5: Internal Flow and Engineering Applications	8
Laminar and turbulent flow; Reynolds number; Flow through pipes; Major and minor losses; Boundary layer concept; Introduction to pumps and turbines; Hydraulic systems in mechatronics; Industrial thermofluidic applications.	

Course Outcomes:

Course Outcome (CO)		Bloom's Level
At the end of this course, the students will be able to:		
CO 1	Understand fundamental concepts of thermodynamic systems and energy interactions.	K2
CO 2	Apply laws of thermodynamics to engineering devices and processes.	K3
CO 3	Study properties and behavior of fluids under static conditions.	K3
CO 4	Study fluid flow equations to solve engineering problems involving pipes and fluid machinery.	K3

CO 5	Study thermofluidic systems used in mechanical and mechatronic applications.	K3
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K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate, K6 – Create

Text Books:

- P.K. Nag — Engineering Thermodynamics, Tata McGraw-Hill (5th Ed.)
- Y.A. Cengel & M.A. Boles — Thermodynamics: An Engineering Approach, McGraw-Hill (8th Ed.)
- C.P. Arora — Thermodynamics, Tata McGraw-Hill
- P. N. Modi, S. M. Seth — Fluid Mechanics and Hydraulic Machinery, Standard Book House, 10th Ed., 1991.
- Robert W. Fox, Alan T. McDonald — Introduction to Fluid Mechanics, John Wiley and Sons, 5th Ed.

Reference Books:

- G.F.C. Rogers & Y.R. Mayhew — Engineering Thermodynamics: Work and Heat Transfer, Longman
- Dr. R. K. Bansal — Fluid Mechanics and Hydraulic Machines, Laxmi Publication, Delhi, 2005
- V. L. Streeter, K. W. Bedford and E. B. Wylie — Fluid Dynamics, Tata McGraw-Hill, 9th Ed., 1998.
- S. K. Som, G. Biswas — Introduction to Fluid Mechanics and Fluid Machines, Tata McGraw Hill, 2nd Ed., 2003.

CME 302: MATERIALS SCIENCE & METALLURGY

Course Objectives:

The objective of this course is to impart knowledge of material structure, properties, behavior and testing techniques relevant to mechanical and mechatronic engineering applications.

1. Understand crystal structures, material defects and their effect on properties of engineering materials.
2. Apply stress-strain analysis and mechanical testing methods to evaluate material behavior.
3. Analyze phase diagrams to understand engineering alloys and microstructural transformations.
4. Explain heat treatment and surface hardening processes for improving component performance.
5. Apply metallography and material selection principles for manufacturing and mechatronic applications.

Syllabus:

Content	Contact Hours
Unit-1: Crystal Structure and Material Defects	8
Classification of materials (Metals, Ceramics, Polymers and composites); Crystalline and amorphous materials; unit cell, BCC, FCC and HCP structures; Coordination number and atomic packing factor; Miller indices and crystallographic directions; Introduction to crystal defects: point defects, dislocations, grain boundaries.	
Unit-2: Mechanical Properties and Material Testing	8
Stress-strain behaviour of materials; Elastic and Plastic deformation; Mechanical properties: strength, hardness, toughness, ductility and stiffness; Mechanical testing; Introduction to non-destructive testing.	
Unit-3: Engineering Alloys and Phase Diagrams	8
Introduction to phase diagrams and their importance; Gibbs phase rule; Cooling curves; Lever rule; Iron-Iron carbide diagram; Phases in steel; Pearlite, Bainite and Martensite; TTT diagrams; Classification and applications of steels, stainless steels and aluminium alloys.	
Unit-4: Heat Treatment and Surface Engineering	8
Objectives of heat treatment; annealing, normalizing, hardening and tempering; hardenability and Jominy test; surface hardening methods: carburizing, nitriding, flame hardening and induction hardening; applications in gears, shafts, tools and machine components.	
Unit-5: Materials for Additive Manufacturing and Metallography	8
Thermoplastics (PLA, ABS, Nylon etc.); photopolymers; metal powders; introduction to metallography; specimen preparation and etching; optical microscope and microstructure observation; industrial applications.	

Course Outcomes:

Course Outcome (CO)		Bloom's Level
At the end of this course, the students will be able to:		
CO 1	Explain crystal structures, material defects and their influence on engineering material properties used in mechatronics systems.	K2

CO 2	Interpret stress-strain behaviour and evaluate mechanical properties of materials using standard testing methods.	K3
CO 3	Analyze phase diagrams and identify important engineering alloys and microstructures for industrial applications.	K3
CO 4	Explain heat treatment and surface hardening processes for improving performance of machine components.	K2
CO 5	Apply basic metallography and material selection concepts for mechatronics and manufacturing applications.	K3

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate, K6 – Create

Text Books:

- W.D. Callister Jr. & D.G. Rethwisch — Materials Science and Engineering: An Introduction, Wiley (10th Ed.)
- V. Raghavan — Physical Metallurgy: Principles and Practice, Prentice Hall India (5th Ed.)
- Avner S.H. — Introduction to Physical Metallurgy, Tata McGraw-Hill

Reference Books:

- M.F. Ashby & D.R.H. Jones — Engineering Materials 1 & 2, Elsevier (4th Ed.)
- R.E. Reed-Hill & R. Abbaschian — Physical Metallurgy Principles, Thomson Learning (3rd Ed.)
- Higgins R.A. — Engineering Metallurgy, ELBS (6th Ed.)

CME 303: INTRODUCTION TO ROBOTICS

Course Objectives:

The objective of this course is to introduce students to the fundamental concepts of robotics including robot mechanisms, perception, control and industrial applications.

1. Understand the history, classifications and anatomy of robotic systems.
2. Study robot configurations, joints, links, and basic kinematic concepts.
3. Apply knowledge of sensors and perception systems to robotic design.
4. Understand robot programming methods and control techniques.
5. Apply robotics concepts in industrial, service and collaborative robot applications.

Syllabus:

Content	Contact Hours
Unit-1: Fundamentals of Robotics	8
Introduction to Robotics, History and Evolution of Robots, Advantages and Limitations of Robots, Classification of Robots, Robot Anatomy and Components, Degrees of Freedom (DOF), Robot Coordinate Systems, Applications of Robotics in Industry and Daily Life.	
Unit-2: Robot Mechanisms and Kinematics	8
Types of Robot Configurations: Cartesian, Cylindrical, Spherical, SCARA, Articulated Robots; Robot Joints and Links; Basic Concepts of Forward and Inverse Kinematics; End Effectors and Grippers; Robot Workspace and Reachability.	
Unit-3: Robot Perception and Sensor Integration	8
Sensor Interfacing with Microcontrollers; Encoders as Position Sensors; Inertial Sensors in Robotics; Camera as a Sensor: Conceptual Overview; Sensor Selection and Placement for Robotic Tasks.	
Unit-4: Robot Control and Programming	8
Introduction to Robot Control Systems: Open Loop and Closed Loop Control, Basic Motion Control; Robot Programming Methods: Lead-through Programming, Online and Offline Programming; Introduction to Programming Languages for Robots; Basics of Artificial Intelligence in Robotics.	
Unit-5: Industrial Robotics and Applications	8
Industrial Robotics; Material Handling Applications; Welding and Painting Robots; Assembly Robots; Mobile Robots and Service Robots; Collaborative Robots (Cobots); Future Trends in Robotics; Safety Standards in Robotics.	

Course Outcomes:

Course Outcome (CO)		Bloom's Level
At the end of this course, the students will be able to:		
CO 1	Describe the fundamentals, history, and classifications of robots.	K1
CO 2	Explain robot components, kinematics, and drive systems.	K2
CO 3	Apply knowledge of robot perception — interfacing sensors, interpreting data, and designing sensor systems for robotic tasks.	K3
CO 4	Understand basic robot programming and control techniques.	K2
CO 5	Apply robotic concepts in industrial and service applications.	K3

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate, K6 – Create

Text Books:

- Introduction to Robotics: Mechanics and Control — John J. Craig, Pearson (3rd Ed.)
- Robotics for Engineers — Yoram Koren, McGraw-Hill
- Industrial Robotics — Mikell P. Groover, McGraw-Hill

Reference Books:

- Robotics: Control, Sensing, Vision and Intelligence — K. S. Fu, McGraw-Hill
- Fundamentals of Robotics — Robert J. Schilling, Prentice Hall
- Robot Modeling and Control — Mark W. Spong, Seth Hutchinson, John Wiley & Sons

CME 304: ANALOG AND DIGITAL ELECTRONICS

Course Objectives:

The objective of this course is to provide students with a comprehensive understanding of analog and digital electronic circuits for engineering applications.

1. Understand operational amplifier characteristics and design Op-Amp application circuits.
2. Apply Boolean algebra and Karnaugh map techniques to simplify and implement combinational logic circuits.
3. Design data-processing circuits including multiplexers, decoders, encoders and arithmetic units.
4. Analyze and design sequential circuits using flip-flops, registers and shift registers.
5. Design and verify digital circuits using Hardware Description Languages (HDL).

Syllabus:

Content	Contact Hours
Unit-1: Introduction to Analog Circuits	8
Introduction to Operational Amplifier: Ideal vs. practical Op Amp, Performance Parameters, Operational Amplifier Application Circuits: Peak Detector Circuit, Comparator, Active Filters, Non-Linear Amplifier, Relaxation Oscillator, Current-To-Voltage Converter, Voltage to Current Converter.	
Unit-2: Boolean Algebra and Logic Design	8
Review of Basic Logic gates, Sum-of-Products Method, Truth Table to Karnaugh Map, Pairs Quads, and Octets, Karnaugh Simplifications, Don't care Conditions, Product-of-sums Method, Product-of-sums simplifications, Simplification by Quine-McClusky Method, Hazards and Hazard covers.	
Unit-3: Data-Processing Circuits	8
Adder, Subtractor, Multiplexers, Demultiplexers, 1:16 Decoder, BCD to Decimal Decoders, Seven Segment Decoders, Encoders, Exclusive-OR Gates, Parity Generators and Checkers, Magnitude Comparator, Programmable Array Logic, Programmable Logic Arrays, Arithmetic Building Blocks, Arithmetic Logic Module.	
Unit-4: Sequential Logic Circuits	8
FLIP-FLOP Timing, JK Master-slave FLIP-FLOP, Switch Contact Bounce Circuits, Various Representation of FLIP-FLOPs, Registers: Types of Registers, Serial In-Serial Out, Serial In-Parallel Out, Parallel In-Serial Out, Parallel In-Parallel Out, Universal Shift Register, Applications of Shift Registers.	
Unit-5: Introduction to Hardware Description Languages (HDL)	8
Introduction to HDL, Design flow using HDL (VHDL or Verilog), Behavioral, Dataflow, and Structural modeling, Simulation and verification of digital designs, Introduction to synthesis and implementation tools, Implementation of combinational circuits: Adder, subtractor and MUX, Implementation of sequential circuits: Latch, flip-flops and counters.	

Course Outcomes:

Course Outcome (CO)	Bloom's Level
At the end of this course, the students will be able to:	
CO 1 Analyse ideal and practical Op-Amp characteristics and design Op-Amp based application circuits.	K4
CO 2 Simplify Boolean expressions using Karnaugh Maps and Quine-	K2

	McClusky method and implement combinational logic circuits.	
CO 3	Design and implement data-processing circuits such as multiplexers, decoders, encoders and arithmetic logic units.	K3
CO 4	Analyse and design sequential circuits using flip-flops, registers and shift registers for specific applications.	K4
CO 5	Design digital circuits and analyse using HDL techniques with respect to accuracy and resolution.	K4

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate, K6 – Create

Text Books:

- M. Morris Mano and M. D. Ciletti — Digital Design, Pearson Education.
- S. Salivahanan and S. Arivazhagan — Digital Circuits and Design, 3rd Ed., 2009.

Reference Books:

- R.P. Jain — Modern Digital Electronics, Tata McGraw Hill, 4th Ed., 2009.
- Anand Kumar — Fundamental of Digital Circuits, PHI, 4th Ed., 2018.
- W.H. Gothmann — Digital Electronics: An Introduction to Theory and Practice, PHI, 2nd Ed., 2006.
- D.V. Hall — Digital Circuits and Systems, Tata McGraw Hill.

CME 351: THERMODYNAMICS AND FLUID MECHANICS LAB

Course Objectives:

The objective of this laboratory course is to provide hands-on experience with thermodynamics and fluid mechanics experiments, enabling students to verify theoretical concepts through practical investigation.

1. Verify fundamental laws of thermodynamics through controlled laboratory experiments.
2. Measure and analyze fluid properties and pressure distribution using standard experimental setups.
3. Evaluate performance of thermofluid systems such as pumps, heat engines and pipe flow networks.
4. Develop skills in observation, data recording and scientific report writing for engineering experiments.

List of Experiments (Indicative & not limited to):

1. Verification of Zeroth Law of Thermodynamics using temperature measuring devices
2. Determination of specific heat of a substance using calorimeter
3. Study and performance analysis of a simple heat engine test rig
4. Determination of entropy change during heating/cooling process
5. Measurement of fluid properties: viscosity and surface tension
6. Calibration and use of pressure measuring devices (manometer and pressure gauge)
7. Verification of Bernoulli's theorem
8. Verification of continuity equation using flow apparatus
9. Determination of friction losses in pipe flow and evaluation of Reynolds number
10. Study and performance characteristics of centrifugal pump/hydraulic system
11. Study and performance characteristics of reciprocating pump/hydraulic system

Course Outcomes:

Course Outcome (CO)		Bloom's Level
At the end of this course, the students will be able to:		
CO 1	Explain and apply the fundamental concepts of thermodynamics, including laws of thermodynamics, heat transfer, entropy, and thermodynamic processes through laboratory experiments.	K3
CO 2	Measure and evaluate fluid properties, pressure distribution, and fluid flow behavior using standard fluid mechanics experimental setups.	K3
CO 3	Analyze engineering thermofluid systems such as pipe flow networks, pumps, hydraulic systems, heat engines, and refrigeration systems for industrial and manufacturing applications.	K3

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate, K6 – Create

CME 353: ROBOTICS LAB

Course Objectives:

The objective of this laboratory course is to provide practical exposure to robotic systems, enabling students to understand robot anatomy, kinematics, sensor integration, programming and control through hands-on experimentation.

1. Identify and describe components of robotic systems through physical inspection and assembly.
2. Perform forward and inverse kinematics calculations and verify through robot motion experiments.
3. Interface sensors including encoders and IMUs with microcontrollers for robotic perception tasks.
4. Program and control articulated robotic arms for pick-and-place and curve tracing applications.
5. Implement closed-loop PID control for mobile robot navigation tasks.

List of Experiments (Indicative & not limited to):

1. Robot Types & Component Identification
2. Coordinate Systems & Workspace Mapping
3. Forward Kinematics – 2-DOF Planar Arm
4. Inverse Kinematics – Geometric Method
5. Encoder Interfacing with Microcontroller
6. IMU (MPU-6050) for Orientation Measurement
7. Open-Loop vs Closed-Loop (PID) Motor Control
8. Lead-Through (Teach & Playback) Programming
9. Trajectory Planning: Joint Space vs Cartesian
10. Mobile Robot Line Following with PID
11. Cobot Safety & Force Limiting
12. ROS Publisher-Subscriber Communication
13. Assembly of different types of robotic systems
14. Programming and control of Articulated robotic arm
15. Experiment on picking and placing of objects using robotic cell
16. Experiment on curve tracing using robotic cell
17. Experiment on robotic welding
18. Experiment on gripping the different types of objects using articulated robotic arm

Course Outcomes:

Course Outcome (CO)		Bloom's Level
At the end of this course, the students will be able to:		
CO 1	Identify different types of robots, their components, coordinate systems, and workspace envelope.	K2
CO 2	Derive and apply forward and inverse kinematics for a 2-DOF planar arm using geometric methods, and plan joint-space/Cartesian trajectories.	K3
CO 3	Interface encoders and IMU (MPU-6050) with a microcontroller, and implement open-loop and closed-loop (PID) motor control for orientation and motion feedback.	K3
CO 4	Program robotic systems using lead-through (teach & playback) methods and ROS publisher–subscriber communication, including PID-	K4

	based line following for mobile robots and cobot safety/force-limiting principles.	
CO 5	Assemble robotic systems and operate an articulated robotic arm/robotic cell to perform pick-and-place, curve tracing, welding, and gripping of different object types.	K4

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate, K6 – Create

CME 354: ANALOG AND DIGITAL ELECTRONICS LAB

Course Objectives:

The objective of this laboratory course is to provide practical experience in designing, constructing and testing analog and digital electronic circuits, reinforcing concepts covered in the theory course.

1. Design and test Op-Amp based analog circuits including inverting, non-inverting and voltage follower configurations.
2. Implement and verify truth tables of basic and universal logic gates.
3. Design and test combinational circuits including adders, subtractors, encoders, decoders and multiplexers.
4. Design and verify sequential digital circuits using flip-flops.
5. Use simulation tools such as Proteus or NI Multisim for digital circuit design and verification.

List of Experiments (Indicative & not limited to):

1. Design, construction and testing of inverting amplifier using Op-Amp
2. Design, construction and testing of non-inverting amplifier using Op-Amp
3. Design, construction and testing of voltage follower amplifier using Op-Amp
4. Design and implement Basic gates
5. Design and implement Universal gates
6. Design and implement Half Adder and Full Adder using basic gates
7. Design and implement Half Subtractor and Full Subtractor using basic gates
8. Design and Verification of truth table of J-K Flip Flop
9. Design and Verification of truth table of D Flip Flop
10. Design and Implementation of encoder using logic gates
11. Design and Implementation of 4x1 multiplexer using logic gates
12. Design and Implementation of decoder using logic gates

Course Outcomes:

Course Outcome (CO)		Bloom's Level
At the end of this course, the students will be able to:		
CO 1	Design and test Op-Amp based analog circuits (inverting, non-inverting, voltage follower) in the laboratory.	K3
CO 2	Implement and verify the truth tables of basic and universal logic gates and combinational circuits (Adder, Subtractor).	K3
CO 3	Design and implement sequential digital circuits including Flip-Flops, Encoders, Multiplexers and Decoders.	K3

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate, K6 – Create

CVA 301: CONSTITUTION OF INDIA

Course Objectives:

The objective of this course is to familiarize students with the fundamental principles of the Indian Constitution and its significance in democratic governance.

1. Understand the historical background, making and salient features of the Indian Constitution.
2. Study Fundamental Rights, Fundamental Duties and Directive Principles of State Policy.
3. Understand the structure and functioning of the Parliamentary and Federal systems of India.
4. Study Constitutional amendments, Emergency provisions and Local Self Government schemes.
5. Appreciate the role of the Constitution in shaping Indian democracy and governance.

Syllabus:

Content	Contact Hours
Unit-1: Introduction and Basic Information about Indian Constitution	8
Meaning of the constitution law and constitutionalism, Historical Background of the Constituent Assembly, Government of India Act of 1935 and Indian Independence Act of 1947, Enforcement of the Constitution, Indian Constitution and its Salient Features, The Preamble of the Constitution, Fundamental Rights, Fundamental Duties, Directive Principles of State Policy, Parliamentary System, Federal System, Centre-State Relations, Amendment of the Constitutional Powers and Procedure, Emergency Provisions: National Emergency, President Rule, Financial Emergency, and Local Self Government- Constitutional Scheme in India.	
Unit-2: Union Executive and State Executive	8
Powers of Indian Parliament; Functions of Rajya Sabha and Lok Sabha; Powers and Functions of the President; Comparison of powers of Indian President with the United States; Powers and Functions of the Prime Minister; Judiciary — Independence of the Supreme Court, Appointment of Judges, Judicial Review, Public Interest Litigation, Judicial Activism, Lokpal, Lok-Ayukta; State Executives — Powers and Functions of the Governor, Chief Minister, State Cabinet, State Legislature, High Court and Subordinate Courts.	
Unit-3: Introduction and Basic Information about Legal System	8
The Legal System: Sources of Law and the Court Structure; Enacted law — Acts of Parliament (primary legislation); Common Law or Case law; The Court System in India (District Court, Consumer Forum, Tribunals, High Courts, Supreme Court); Arbitration; Contract law, Tort, Law at workplace.	
Unit-4: Intellectual Property Laws and Regulation to Information	8
Intellectual Property Laws- Introduction, Legal Aspects of Patents, Filing of Patent Applications, Rights from Patents, Infringement of Patents, Copyright and its Ownership, Infringement of Copyright, Civil Remedies for Infringement, Regulation to Information- Introduction, Right to Information Act, 2005, Information Technology Act, 2000, Electronic Governance, Secure Electronic Records and Digital Signatures, Digital Signature Certificates, Cyber Regulations Appellate Tribunal, Offences, Limitations of the Information Technology Act.	
Unit-5: Business Organizations and E-Governance	8
Sole Traders, Partnerships: Companies: The Company's Act: Introduction, Formation of a Company, Memorandum of Association, Articles of Association, Prospectus,	

Shares, Directors, General Meetings and Proceedings, Auditor, Winding up. E-Governance and role of engineers in E-Governance, Need for reformed engineering serving at the Union and State level, Role of I.T. professionals in Judiciary, Problem of Alienation and Secessionism in few states creating hurdles in Industrial development.	
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Course Outcomes:

Course Outcome (CO)		Bloom's Level
At the end of this course, the students will be able to:		
CO 1	Understand the historical background, making, and salient features of the Indian Constitution including Fundamental Rights and Duties.	K2
CO 2	Analyze the structure and functions of the Union and State Executives, and the role of the Judiciary in Indian governance.	K4
CO 3	Understand the legal system, sources of law, court structure, and alternative dispute resolution mechanisms.	K2
CO 4	Evaluate intellectual property laws and regulations governing information technology and digital governance.	K5
CO 5	Apply knowledge of business organizations and e-governance to understand the role of engineers in democratic society.	K3

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate, K6 – Create

Text Books:

- Brij Kishore Sharma — Introduction to the Indian Constitution, PHI, New Delhi, latest edition.
- Granville Austin — The Indian Constitution: Corner stone of a Nation, Oxford Clarendon Press, 1966.
- Subhash C. Kashyap — Our Constitution: An Introduction to India's Constitution and Constitutional Law, NBT, 2018.
- P. M. Bakshi — The Constitution of India, Latest Edition, Universal Law Publishing.

Reference Books:

- V. K. Ahuja — Law Relating to Intellectual Property Rights, 2007.
- Suresh T. Viswanathan — The Indian Cyber Laws, Bharat Law House, New Delhi.
- P. Narayan — Intellectual Property Law, Eastern Law House, New Delhi.

B.Tech. Mechanical & Mechatronics Engineering (Additive Manufacturing)

Second Year, Semester-IV

CME 401: MECHANICS OF SOLIDS

Course Objectives:

The objective of this course is to provide students with a thorough understanding of stress, strain and deformation behaviour of solid bodies under various loading conditions.

1. Understand concepts of stress, strain and material properties under different loading conditions.
2. Analyze beams, shafts and columns under bending, torsion and axial loads.
3. Apply theories of failure to design and evaluate machine and structural components.
4. Study deflection of beams and statically indeterminate structures.
5. Apply principles of mechanics of solids to practical engineering problems.

Syllabus:

Content	Contact Hours
Unit-1: Principal Stresses and Strains	8
Uniaxial stress and simple shear; General state of stress for 2-D element; Principal stresses and Principal planes; Principal strains and Shear strains; Strain rosettes; Theories of failure.	
Unit-2: Strain Energy, Resilience and Combined Stresses	8
Load-deflection diagram; Strain energy; Proof resilience; Stresses due to gradual, sudden and impact loading; Shear resilience; Combined axial and flexural loads; Eccentrically applied load on columns; Columns & Struts: Concept of short and long columns; Euler's formula and its limitations.	
Unit-3: Stresses in Beams	8
Moment of inertia of different sections; Theory of simple bending; Derivation of flexural formula; Bending and Shearing stresses in a beam; Economic sections; Horizontal and vertical shear stress; Distribution shear stress for rectangular, solid circular, I-section and other sections; Design for flexure and shear; Torsion: Introduction, derivation of torsion formula, torsion of circular shafts.	
Unit-4: Shear Force and Bending Moment Diagram	8
Introduction to different types of beams, supports & loads; Concept and definition of shear force and bending moment in determinate beams due to concentrated loads, UDL, UVL and couple; Relation between SF, BM and intensity of loading; Construction of SFD and BMD for cantilever, simple and compound beams; Point of contra flexure. Construction of BMD and load diagram from SFD, Construction of load diagram and SFD from BMD.	
Unit-5: Deflection of Beams	8
Differential equation of deflected beam; Slope and deflection at a point; Double integration method for determinate beams; Mohr's theorems (area moment method); Moment diagram by parts; Deflection of cantilever beams and simply supported beams.	

Course Outcomes:

Course Outcome (CO)		Bloom's Level
At the end of this course, the students will be able to:		
CO 1	Determine principal stress, strain, principal stresses, principal planes, strain rosettes and theories of failure for 2-D stress systems.	K2
CO 2	Study strain energy, resilience and behaviour of columns under various loading conditions including eccentric loading.	K2
CO 3	Describe the theory of simple bending and torsion, and interpret bending and shear stress distribution across standard cross-sections.	K2
CO 4	Construct Shear Force Diagrams (SFD) and Bending Moment Diagrams (BMD) for cantilever, simply supported and compound beams under concentrated loads, UDL, UVL and couples.	K3
CO 5	Apply double integration method, Mohr's area-moment theorems and method of superposition to compute slope and deflection at critical points in determinate beams.	K3

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate, K6 – Create

Text Books:

- R.K. Bansal — Strength of Materials, Laxmi Publications
- E.P. Popov — Engineering Mechanics of Solids, Pearson (2nd Ed.)
- Beer & Johnston — Mechanics of Materials, McGraw-Hill (7th Ed.)

Reference Books:

- S.P. Timoshenko & J.N. Goodier — Theory of Elasticity, McGraw-Hill (3rd Ed.)
- Sadhu Singh — Strength of Materials, Khanna Publishers
- R.K. Rajput — Strength of Materials, S. Chand & Company

CME 402: MANUFACTURING SCIENCE

Course Objectives:

The objective of this course is to introduce students to the fundamental principles and processes of manufacturing science including casting, forming, machining and modern manufacturing systems.

1. Understand the principles and process parameters of casting and solidification.
2. Study forming processes including forging, rolling, extrusion and sheet metal operations.
3. Apply knowledge of machining processes, tool geometry and cutting theory to manufacturing problems.
4. Understand welding processes, joint design and quality assessment.
5. Analyze plant layout, product design and sustainable manufacturing principles.

Syllabus:

Content	Contact Hours
Unit-1: Plant Layout & Manufacturing Concepts	8
Factors in Plant Layout, Principles and Objectives of Plant Layout, Types of Plant Layout, Comparison of Different Layouts, Manufacturing — Product Design and Concurrent Engineering, Sustainable Manufacturing.	
Unit-2: Casting Processes	8
Metal Casting: Patterns, Pattern Material, Types of Pattern, Pattern Allowances, Moulding, Mould Materials, Moulding Sand Constituents, Types of Mould, Moulding Methods, Core Types, Core Making, Melting Equipment, Casting Processes Classification, Permanent Mould and Die Casting, Centrifugal Casting, Investment Casting, Continuous Casting, Defects in Castings.	
Unit-3: Metal Forming & Plastic Processing	8
Hot Working, Hot Working Processes, Hot Rolling, Hot Forging, Hot Extrusion, Hot Drawing, Cold Working, Cold Working Processes, Cold Rolling, Cold Forging, Cold Extrusion, Cold Bending, Cold Drawing; Compression Moulding, Transfer Moulding, Injection Moulding, Blow Moulding, Extrusion, Thermoforming.	
Unit-4: Welding Processes	8
Welding Classification, Oxy Acetylene Gas Welding, Electric Arc Welding Principle, Carbon Arc Welding, Flux Shielded Metal Arc Welding, TIG, MIG, Resistance Welding Principle, Spot and Seam Welding, Friction Welding, Ultrasonic Welding, Thermit Welding, Laser Beam and Electron Beam Welding, Welding Defects.	
Unit-5: Machining Processes	8
Traditional Machining: Cutting Tools, Cutting Forces, Mechanism of Chip Formation, Lathe, Types of Lathe, Principal Parts of Lathe, Lathe Operations, Drilling, Shaper, Planer, Slotter, Milling, Grinding. Non-Traditional Machining: Electrochemical Machining (ECM), Electrical Discharge Machining (EDM), Ultrasonic Machining (USM) and Laser Beam Machining (LBM).	

Course Outcomes:

Course Outcome (CO)	Bloom's Level
At the end of this course, the students will be able to:	
CO 1 Understand about principles and objectives of plant layouts and describe the role of concurrent engineering and sustainable manufacturing in modern production systems.	K2

CO 2	Understand pattern-making, moulding and core-making processes, classify casting methods and identify common casting defects.	K2
CO 3	Study hot and cold metal forming processes and describe polymer processing methods such as injection moulding, blow moulding and thermoforming.	K2
CO 4	Apply knowledge of welding processes to select an appropriate method for a given joint configuration, material and service condition, and identify common welding defects.	K3
CO 5	Apply understanding of traditional and non-traditional machining processes to select a suitable process (ECM, EDM, USM or LBM) for a given work material and geometry requirement.	K3

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate, K6 – Create

Text Books:

- Comprehensive Workshop Technology (Manufacturing Processes) — S. K. Garg, Laxmi Publications.
- Manufacturing Technology (Volume 2) Foundry, Forging and Welding — P.N. Rao, 4th Ed., Tata McGraw Hill Education.
- A Textbook of Manufacturing Technology: Manufacturing Processes — R. K. Rajput, Laxmi Publications.
- Manufacturing Engineering and Technology — SeropeKalpakjian; Steven R. Schmid, 6th Ed., Prentice Hall.

Reference Books:

- Manufacturing Processes — H. N. Gupta, New Age Publishers, 2012.
- Fundamentals of Modern Manufacturing: Materials, Processes and Systems — Mikell P. Groover, Wiley India, 2012.
- Production Technology — R.K. Jain, Khanna Publishers.

CME 403: COMPUTER-AIDED DESIGN AND ADDITIVE MANUFACTURING

Course Objectives:

The objective of this course is to introduce students to computer-aided design principles and additive manufacturing processes for modern engineering product development.

1. Understand principles of computer-aided design and geometric modeling techniques.
2. Apply CAD software tools for 2D drafting and 3D parametric solid modeling.
3. Study the working principles, process parameters and applications of various additive manufacturing technologies.
4. Analyze design for additive manufacturing considerations including support structures and build orientation.
5. Apply CAD and AM integration for product development and rapid prototyping.

Syllabus:

Content	Contact Hours
Unit-1: Fundamentals and Geometric Modelling	8
Introduction to Computer-Aided Design (CAD); CAD system components; Coordinate systems and Basic geometric modelling; Comparison of additive and subtractive manufacturing; CAD file formats and data exchange.	
Unit-2: 3D Modelling and Product Design	8
Sketching and feature-based solid modelling; Parametric modelling concepts; Basic assembly modelling; Engineering drawing; Dimensions and annotations; Introduction to design considerations for Additive Manufacturing (AM).	
Unit-3: CAD-CAE Integration & Reverse Engineering	8
Introduction to CAE and Finite Element Analysis (FEA); Basic meshing strategies and mesh quality criteria; Static structural analysis: Boundary conditions; CAD-CAM-CAE integration; Reverse engineering concepts; 3D scanning technologies; Point cloud to CAD workflow (scan-to-CAD).	
Unit-4: Additive Manufacturing — Processes & Materials	8
Introduction to Additive Manufacturing; Generic AM process chain; Advantages, limitations and industrial applications. Polymer AM: FDM/FFF: Process parameters, supports, infill; SLA/DLP — photopolymer chemistry, layer resolution. Metal AM: SLM/LPBF; DED and WAAM; Materials in AM; Surface quality, dimensional accuracy and common AM defects.	
Unit-5: Design for AM (DfAM) & Emerging Applications	8
Design considerations specific to AM; Design for Additive Manufacturing (DfAM) methodology; Topology optimisation and generative design concepts; Lightweight structures and lattice design (TPMS, strut-based lattices); Quality inspection in AM; AM in smart manufacturing; Future trends.	

Course Outcomes:

Course Outcome (CO)		Bloom's Level
At the end of this course, the students will be able to:		
CO 1	Explain the components of a CAD system, differentiate geometric modelling techniques (wireframe, surface, solid) and describe standard CAD data exchange formats.	K2
CO 2	Apply parametric and feature-based modelling tools to create solid models, assemblies and fully annotated engineering drawings.	K3

CO 3	Apply CAD-CAE integration tools to set up a basic FEA model, interpret static analysis results and execute a reverse-engineering scan-to-CAD workflow.	K3
CO 4	Explain the AM process chain, compare polymer and metal AM processes and identify the material properties and common defects associated with each.	K2
CO 5	Apply DfAM principles to design topology-optimised and lattice structures, and evaluate AM part quality using appropriate inspection techniques.	K3

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate, K6 – Create

Text Books:

- CAD/CAM: Principles, Practice and Manufacturing Management — McMahon & Browne (Pearson)
- CAD/CAM: Computer-Aided Design and Manufacturing — Groover&Zimmers (Prentice Hall)
- Mastering CAD/CAM — Ibrahim Zeid (Tata McGraw-Hill)
- Additive Manufacturing Technologies — Gibson, Rosen, Stucker, Khorasani (Springer, 3rd Ed. 2021)

Reference Books:

- 3D Printing and Additive Manufacturing — Chua & Leong (World Scientific, 5th Ed.)
- Product Design for Manufacture and Assembly — Boothroyd, Dewhurst & Knight (CRC Press, 3rd Ed.)
- Finite Element Procedures — K.J. Bathe (Prentice Hall)

CME 404: INTRODUCTION TO AUTOMATION

Course Objectives:

The objective of this course is to introduce students to the principles and applications of industrial automation including sensors, actuators, PLCs and automated manufacturing systems.

1. Understand the concept and need for automation in modern manufacturing systems.
2. Study types of sensors, actuators and transducers used in automated systems.
3. Apply knowledge of PLCs and ladder logic for programming industrial controllers.
4. Understand material handling systems and automated guided vehicles.
5. Apply automation concepts to design and evaluate automated manufacturing cells.

Syllabus:

Content	Contact Hours
Unit-1: Fundamentals of Automation	8
Definition, history, need and scope; Economy of Scale & Scope; Types of production systems; Hard vs. Soft Automation; Automation Strategies; Components of automation system; Social and environmental effects.	
Unit-2: Signal Conditioning and Data Acquisition Systems	8
Automation Pyramid; DAQ architecture; Signal conditioning: amplification, filtering, A/D & D/A conversion; SCADA architecture and HMI; Industrial protocols: Modbus, PROFIBUS, CAN Bus, Fieldbus; DCS vs. SCADA.	
Unit-3: Programmable Logic Controllers	8
PLC architecture: CPU, I/O modules, scan cycle; Ladder Logic: contacts, coils, timers, counters; FBD and Structured Text; PLC wiring and interfacing; Industrial applications; PAC introduction.	
Unit-4: Process Control & Modelling	8
Self-regulating vs. non-self-regulating processes; Feedback, feedforward and cascade control; Mathematical modelling: thermal, liquid-level, pressure systems; Transfer functions; PID tuning: Ziegler-Nichols, Cohen-Coon.	
Unit-5: Advanced Automation & Industry 4.0	8
Internet of Things (IoT)/Industrial Internet of Things (IIoT): smart sensors, edge computing, cloud; Cyber-Physical Systems and Digital Twin; CIM and FMS; Machine Vision for quality control; Cobots introduction; Industry 4.0 overview.	

Course Outcomes:

Course Outcome (CO)		Bloom's Level
At the end of this course, the students will be able to:		
CO 1	Understand and describe the fundamentals, history, need and scope of automation, types of production systems and components of an automation system.	K1
CO 2	Explain DAQ architecture, signal conditioning techniques and industrial communication protocols used in automated systems.	K2
CO 3	Apply knowledge of PLC architecture and ladder logic programming to design and implement industrial control applications.	K3
CO 4	Analyze process control strategies and apply mathematical modelling and PID tuning techniques to design control systems.	K3
CO 5	Understand advanced automation concepts including IoT/IIoT, Digital Twin, CIM, FMS and Industry 4.0 technologies.	K2

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate, K6 – Create

Text Books:

- Automation, Production Systems, and Computer-Integrated Manufacturing — Mikell P. Groover
- Programmable Logic Controllers — Frank D. Petruzella
- Process Control: Modelling, Design & Simulation — B. W. Bequette, Prentice Hall

Reference Books:

- Industrial Instrumentation and Control — S. K. Singh
- Principles of Industrial Instrumentation — D. Patranabis
- Control Systems Engineering — Norman S. Nise, Wiley

CME 405: INTRODUCTION TO ARTIFICIAL INTELLIGENCE

Course Objectives:

The objective of this course is to introduce students to fundamental concepts of artificial intelligence and machine learning with applications in engineering and mechatronic systems.

1. Understand the history, concepts and problem-solving approaches in artificial intelligence.
2. Study search algorithms, knowledge representation and reasoning techniques.
3. Apply machine learning algorithms to classify and predict outcomes from engineering data.
4. Understand neural networks, deep learning fundamentals and their engineering applications.
5. Apply AI and ML techniques to solve real-world mechatronics and manufacturing problems.

Syllabus:

Content	Contact Hours
Unit-1: Introduction	8
Introduction to AI, the foundations of Artificial Intelligence, history of Artificial Intelligence, AI Agents, structure of Intelligent Agents, nature of environments.	
Unit-2: Problem Solving	8
Problem-Solving agents, searching solutions, various uninformed and informed (Heuristic) search techniques, Constraint Satisfaction Problems.	
Unit-3: Planning	8
Planning — A Simple Planning Agent, Problem Solving to Planning, Planning in Situation Calculus, Basic Representations for Planning, A Partial-Order Planning Example & Algorithm, Planning with Partially Instantiated Operators, Knowledge Engineering for Planning.	
Unit-4: Reasoning	8
Uncertain Knowledge and reasoning: Uncertainty — Acting under Uncertainty, Basic Probability Notation, the Axioms of Probability, Bayes' Rule and Its Use; Probabilistic Reasoning systems — Representing Knowledge in an Uncertain Domain, The Semantics of Belief Networks, Inference in Belief Networks, Inference in Multiply Connected Belief Networks.	
Unit-5: Learning	8
Learning: Learning from Observations — A General Model of Learning Agents, Inductive Learning, Learning Decision Trees, Using Information Theory, Learning General Logical Descriptions, and Computational Learning Theory; Learning in Neural and Belief Networks — Neural Networks, Perceptions, Multilayer Feed-forward Networks, Applications of Neural Networks.	

Course Outcomes:

Course Outcome (CO)	Bloom's Level
At the end of this course, the students will be able to:	
CO 1 Understand the foundations, history and agent-based architecture of Artificial Intelligence systems and the nature of environments.	K2
CO 2 Apply uninformed and informed search algorithms and constraint satisfaction techniques to solve AI problem-solving tasks.	K3

CO 3	Develop AI planning strategies using situation calculus and partial-order planning algorithms.	K3
CO 4	Apply probabilistic reasoning and Bayesian belief networks to handle uncertainty in AI systems.	K3
CO 5	Understand machine learning concepts including decision trees and neural networks and their real-world applications.	K2

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate, K6 – Create

Text Books:

- Stuart J. Russell and Peter Norvig — Artificial Intelligence: A Modern Approach, 3rd Ed., Prentice Hall, 2003.
- Nils J. Nilsson — Artificial Intelligence: A New Synthesis, Morgan Kaufmann Publishers.

Reference Books:

- Patrick H. Winston — Artificial Intelligence, Addison-Wesley, 3rd Ed.
- Elaine Rich, Kevin Knight and Shivashankar B. Nair — Artificial Intelligence, Tata McGraw-Hill, 3rd Ed.
- George F. Luger — Artificial Intelligence: Structures and Strategies for Complex Problem Solving, Pearson, 6th Ed.

CME 451: MATERIALS TESTING LAB

Course Objectives:

The objective of this laboratory course is to provide practical experience in testing mechanical properties of materials using standard destructive and non-destructive testing methods.

List of Experiments (Indicative & not limited to):

1. Strength test of a given mild steel specimen on UTM with full details and stress versus strain plot on the machine.
2. Other tests such as shear, bend tests on UTM.
3. Impact test on impact testing machine (Charpy, Izod or both).
4. Hardness test of given specimen using Rockwell and Vickers/Brinell testing machines — conducted on MS specimen prior and after various heat-treatment processes like Annealing, Normalizing, Hardening etc.
5. Fatigue test on fatigue testing machine.
6. Creep test on creep testing machine.
7. Study of NDT (non-destructive testing) methods like magnetic flaw detector, ultrasonic flaw detector, eddy current testing machine, dye penetrant tests.
8. Modelling of Graphene / any other materials structure using LAMMPS (Molecular Dynamics Simulator) or any other simulation software and studying the mechanical properties.
9. Measurement of mechanical properties (tensile test, compression test, hardness test / flexural test) of specimen made by 3-D printing.

Course Outcomes:

Course Outcome (CO)		Bloom's Level
At the end of this course, the students will be able to:		
CO 1	Students will be able to perform different destructive and non-destructive testing methods to measure various mechanical properties.	K2
CO 2	Students will be able to analyse the effect of different heat-treatment processes on the Hardness.	K4
CO 3	Students will be able to simulate the material using simulating software / measure the mechanical properties of 3-D printed components.	K3

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate, K6 – Create

CME 452: MANUFACTURING AND AUTOMATION LAB

Course Objectives:

The objective of this laboratory course is to provide practical exposure to manufacturing processes, CNC machining and automation systems, enabling students to perform real-world fabrication and automation experiments.

1. Perform sheet metal, forging and welding operations and analyze process parameters and defects.
2. Execute CNC part programs for turning and milling operations using G-code on machines or simulators.
3. Measure surface roughness and tool flank wear and determine Taylor tool life equation constants.
4. Conduct EDM experiments and evaluate the effect of process variables on MRR and surface finish.
5. Perform experiments on automated systems including ASRS, SCARA robots and automated assembly cells.

List of Experiments (Indicative & not limited to):

1. Sheet metal operations: blanking, piercing, bending on press tooling; calculate punch-die clearance and blanking force.
2. Study of Hot Rolling Process and Measurement of Rolling Parameters.
3. Study of Forging Operations and Analysis of Forging Defects.
4. MIG (GMAW) Welding of Mild Steel (real machine or simulator).
5. TIG (GTAW) Welding of Stainless Steel Specimens (real machine or simulator).
6. Simulation of Resistance Spot Welding and Study of Friction & Ultrasonic Welding.
7. Identification of Welding Defects.
8. CNC turning: write and execute part program for step turning and threading on CNC lathe (real machine or simulator).
9. CNC milling: write and execute pocket milling program on vertical machining centre (real machine or simulator).
10. To study the basics of drilling (real machine or simulator).
11. To study the effects of grinding variables on grinding forces, specific energy and surface finish.
12. Surface roughness measurement: measure Ra, Rz on turned, milled, and ground surfaces using a profilometer; compare with ISO surface finish grades.
13. Tool life experiment: measure flank wear (VB) at intervals for HSS and carbide tools in turning; plot VB vs. cutting time; determine n and C in Taylor's equation.
14. To study the effects of EDM variables on MRR, surface finish and TWR.
15. Study and experimentation on ASRS in Automation Lab.
16. Automatic Loading and unloading of workpiece in Automation Lab.
17. Automatic transfer of workpiece.
18. Experiment on SCARA Robotic system.
19. Experiment on Automated Assembly System.
20. Experiment on Automated Inspection and Testing.

Course Outcomes:

Course Outcome (CO)		Bloom's Level
At the end of this course, the students will be able to:		
CO 1	Perform metal forming operations (forging, sheet metal) and evaluate process parameters and force calculations.	K3
CO 2	Execute MIG and TIG welding operations and inspect weld quality for common defects.	K3
CO 3	Write and execute CNC part programs for turning and milling operations using standard G-codes on CNC machines/simulators.	K3
CO 4	Measure surface roughness and tool flank wear and analyse tool life using Taylor's equation constants.	K4

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate, K6 – Create

CME 453: COMPUTER AIDED DESIGN AND ADDITIVE MANUFACTURING LAB

Course Objectives:

The objective of this laboratory course is to provide hands-on experience in computer-aided design and additive manufacturing, enabling students to create solid models, perform FEA and study 3D printing processes.

1. Create 2D drawings and 3D parametric solid models using CAD software tools.
2. Generate engineering drawings with dimensions, tolerances and GD&T annotations.
3. Perform basic FEA including meshing, boundary conditions and static stress analysis.
4. Study FDM, SLA and metal AM processes, optimize slicing parameters and evaluate part quality.
5. Apply DfAM principles to design topology-optimised and lattice structures for additive manufacturing.

List of Experiments (Indicative & not limited to):

1. Introduction to CAD Software: Interface, Coordinate Systems and 2D Geometric Construction.
2. Feature-Based Parametric Solid Modelling: Boss-Extrude, Cut-Extrude, Fillet and Shell.
3. Engineering Drawing Generation: Views, Dimensions, GD&T Annotations and Title Block.
4. FDM/FFF Process: Slicing, Parameter Optimisation and Dimensional Accuracy Study.
5. Multi-Jet Printing (MJP): High-Resolution Prototyping and Process Comparison with FDM.
6. Virtual Study of SLA/DLP and Metal AM Processes (SLM/LPBF and WAAM) Using Simulation.
7. Study of Surface Quality and Defects in AM Parts: Staircase Effect, Warping, Stringing and Porosity.
8. Introduction to FEA: Meshing, Static Stress Analysis, and Result Interpretation in CAD-CAE Environment.
9. Design for Additive Manufacturing (DfAM): Topology Optimisation and Lightweight Lattice Design.
10. DfAM Redesign: Support Minimisation, Build Orientation Study and Wall Thickness Analysis.
11. Quality Inspection of AM Parts: Dimensional Metrology and CMM / GD&T Verification.
12. Industry 4.0 and Digital Manufacturing: CAD-CAM Integration, G-code Generation and CNC Simulation.

Course Outcomes:

Course Outcome (CO)		Bloom's Level
At the end of this course, the students will be able to:		
CO 1	Operate CAD software interface, apply coordinate systems and construct 2D geometric profiles for engineering components.	K3
CO 2	Build feature-based parametric solid models using Boss-Extrude, Cut-Extrude, Fillet and Shell operations and generate multi-view engineering drawings with GD&T annotations.	K3
CO 3	Analyse FDM, MJP, SLA/DLP and metal AM (SLM/LPBF, WAAM) processes, optimise slicing parameters and evaluate surface quality and defects in AM parts.	K4

CO 4	Perform FEA including meshing, boundary condition application and static stress result interpretation in a CAD-CAE environment.	K4
CO 5	Apply DfAM principles to redesign components with topology optimisation, lattice structures, support minimisation, build orientation and dimensional metrology using CMM/GD&T; integrate CAD-CAM for G-code generation.	K5

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate, K6 – Create

CME 456: MINI PROJECT

Course Objectives:

The student is required to undertake a short project in the area relevant to Mechatronics. The objective is to provide hands-on training in the practical aspects of the curriculum and to enhance their research and analytical skills.

CVA 401: UNIVERSAL HUMAN VALUES

Course Objectives:

The objectives of this course are:

1. To help students understand the importance of human values in personal and professional life.
2. To develop a holistic perspective towards life, society, and nature.
3. To enable students to differentiate between values and skills, and understand their role in harmony.
4. To promote ethical and responsible behavior in society.
5. To encourage self-exploration and continuous self-improvement.

Syllabus:

Content	Contact Hours
Unit-1: Introduction to Human Values	8
Need, Basic Guidelines, Content and Process for Value Education; Self-Exploration: What is it? Its content and process; 'Natural Acceptance' and Experiential Validation; Continuous Happiness and Prosperity; Right understanding, Relationship and Physical Facility; Understanding the Human Being as Co-existence of Self (I) and Body.	
Unit-2: Understanding Harmony in the Human Being	8
Harmony in the Human Being: Understanding the needs of Self (I) and Body; Understanding the Body as an Instrument of 'I'; Harmony in 'I' (Self): Thoughts, Behaviour, Work; Harmony of Self with Body; Programs to ensure self-regulation and health.	
Unit-3: Understanding Harmony in the Family and Society	8
Harmony in Family: Trust and Respect as the foundational values; Other feelings: Affection, Care, Guidance, Reverence, Glory, Gratitude, Love; Harmony in Society: Understanding values in human relationships; Justice in human-to-human relationships; Role of family in society.	
Unit-4: Understanding Harmony in Nature and Existence	8
Harmony in Nature: Interconnectedness and mutual fulfillment among four orders of nature (Material, Plant, Animal, Human); Understanding existence as co-existence; Environmental sustainability; Role of human beings in maintaining harmony with nature.	
Unit-5: Implications of Human Values in Professional Ethics	8
Professional Ethics: Natural acceptance of human values; Ethical human conduct; Competence in professional ethics; Holistic technologies, production systems and management models; Case studies on ethical practices; Social responsibility and sustainable development.	

Course Outcomes:

Course Outcome (CO)	Bloom's Level
At the end of this course, the students will be able to:	
CO 1 Understand the significance of human values and self-exploration.	K2
CO 2 Analyze harmony within self and between self and body.	K4
CO 3 Apply values to improve relationships in family and society.	K3
CO 4 Understand harmony in nature and environmental sustainability.	K2

CO 5	Apply ethical principles in professional and social life.	K5
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K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate, K6 – Create

Text Books:

- A Foundation Course in Human Values and Professional Ethics — R.R. Gaur, R. Sangal, G.P. Bagaria

Reference Books:

- Human Values — A.N. Tripathi
- Professional Ethics and Human Values — M. Govindarajan, S. Natarajan, V.S. Senthil Kumar